CS1231S Practice Problems, by TA Ivan

Here is a collection of problems. You may be able to use them as practice for the final exam. Some of the problems might be out of scope, because they were tested in earlier quizzes. But it's still a good idea to try them, because a lot of the course material builds up from earlier parts.

On the other hand, these problems are compiled by me (TA Ivan), not by the professor. They might not be reflective of what you will see in the final exam.

The problems are roughly ordered by what course material you are expected to use, from Handout 1 to Handout 4. The problems are *not* ordered by difficulty. In fact, many of these problems are difficult. They generally ask you to prove something, rather than apply a definition in an example. It is fine to skip a problem and go back to it later, or ask for hints or help.

You can discuss on Telegram for hints and solutions.

1 Logic and set theory

Logic, as in propositions and logical operators, was in the first half of Handout 1. Basic set theory was in the first third of Handout 2. I believe these are out of scope of the final exam, but the proof methods – making use of results you already know, proving set equalities, etc – are useful throughout the entire course.

Problem 1. Using logical equivalences, prove Peirce's law. The following statement is a tautology:

$$((p \to q) \to p) \to p$$

Problem 2. If A, B, C are sets, prove

$$A \cup B = B \cup C = C \cup A \qquad \text{if and only if} \qquad (A \cap B) \cup (B \cap C) \cup (C \cap A) = A \cup B \cup C.$$

2 Proofs

Proofs, in the most fundamental form, were explored in the second half of Handout 1. But there were other results, such as the Well-Ordering Principle (in Handout 2 page 22 (Post-lecture 2d)), that are also relevant. Some more specialized proof methods, such as counting arguments (Tutorial 10), are explored in their own sections later.

Problem 3. A real number r is **rational** if there exist integers p, q where

$$q > 0 \qquad \text{and} \qquad r = \frac{p}{q}.$$

A real number is **irrational** if it is not rational. (Also see Tutorial 8 problem 3.)

Prove $\sqrt{2}$ is an irrational number. (**Hint:** You might want to use the Well-Ordering Principle. See Handout 2 page 22 (Post-lecture 2d); see also Tutorial 5 problem 11 for another instance of WOP.)

Problem 4. Prove there exist two positive real numbers x, y such that x, y are both irrational, but x^y is rational. (**Hint:** One possible approach is to look at $\sqrt{2}^{\sqrt{2}}$. There are two cases.)

Problem 5. Mark any 51 points in the 1×1 unit square. Prove there exist 3 of the points that can be contained inside a circle of radius $1/7$. (**Hint:** You will need the Pigeonhole Principle – see Handout 1 page 21 (Post-lecture 1f) – but in a more general form.)

Problem 6. Prove Bernoulli's inequality. If $n \geq 0$ is a natural number and $x \geq -1$ is a real number, then

$$(1+x)^n \geq 1 + nx$$

Problem 7. Let $a_1, a_2, a_3, \ldots$ be nonnegative real numbers. The **infinite series**

$$\sum_{n=1}^{\infty} a_n$$

is said to **converge** if there exists a real number r such that, for all positive integer n,

$$a_1 + a_2 + \ldots + a_n \leq r.$$

Otherwise it **diverges**.

For each of the following infinite series, determine whether it converges or diverges. Prove your answers.

$$\text{(a)} \sum_{n=1}^{\infty} \frac{1}{2^n} \qquad \text{(b)} \sum_{n=1}^{\infty} 1 \qquad \text{(c)} \sum_{n=1}^{\infty} \frac{1}{n}$$

(**Hint:** The idea for (c) is incredibly difficult to find. Prove the sum of the first 2^n terms is $\geq (n+2)/2$.)

Problem 8. A Boolean formula φ is made of only propositions, brackets, and the implication operator $\to$. It does not contain any other operator; in particular, it does not have negation, and it does not have the statements True or False.

Prove φ is satisfiable, i.e. there is an assignment to its propositions so that it evaluates to True.

3 Relations and functions

Relations were explored in the middle third of Handout 2, and functions were explored in the last third of Handout 2. A few aspects of relations were discussed in Handout 4 about graphs. These are about relations and functions in general, as their own objects, rather than their use in other parts of the course.

Problem 9. Let R be a binary relation on a set A. We say R is **quasi-reflexive** if whenever xRy, we have xRx and yRy.

Prove that if R is symmetric and transitive, then R is quasi-reflexive.

Problem 10. Let R be a binary relation on a set A. Suppose that R is both an equivalence relation and a partial order.

(a) Prove R is the identity relation

$$R = \{(x, x) \mid x \in A\}.$$

(b) Suppose furthermore that R is a total order. Prove A is finite and $|A| \leq 1$.

Problem 11. Let R be a binary relation from X to Y. For $x \in X$, define the **image of x**, denoted $R^{\rightarrow}(x)$, as

$$R^{\rightarrow}(x) = \{y \in Y \mid xRy\}.$$

Let R be a binary relation on a set A. Suppose that R is reflexive. Furthermore, suppose R satisfies the following property: for any two elements $x, y \in A$,

$$R^{\rightarrow}(x) = R^{\rightarrow}(y) \qquad \text{or} \qquad R^{\rightarrow}(x) \cap R^{\rightarrow}(y) = \varnothing.$$

Prove R is an equivalence relation.

Problem 12. Let R be a binary relation on a set A. Suppose R is a function; that is, for all $x \in A$, there exists exactly one $y \in A$ such that xRy. Prove each of the following statements:

(a) R is reflexive if and only if $R = i_A$ is the identity function.

(b) R is symmetric if and only if $R \circ R = i_A$.

(c) R is transitive if and only if $R \circ R = R$.

Let $f : X \to X$ be a function. For positive integer n, the **n-th iterate of f**, denoted f^n, is defined recursively as follows:

$$f^n = \begin{cases} f & \text{if } n = 1 \\ f \circ f^{n-1} & \text{if } n \geq 2 \end{cases}$$

(Also see Handout 4 page 3 (Post-lecture 3f).)

Problem 13. Let $f : X \to X$ be a function. Suppose there exists some positive integer n such that f^n is bijective. Prove f is bijective.

Problem 14. Let X be a finite set and $f : X \to X$ be a function. Prove for all $x \in X$, there exist two *distinct* positive integers m, n such that $f^m(x) = f^n(x)$.

4 Cardinality and countability

Cardinality of sets was explored in the first half of Handout 3. This is about finite and infinite sets, and about countable and uncountable sets.

Problem 15. Let A be a set. Define $\mathcal{P}^{\text{fin}}(A)$ as the set of all finite subsets of A, and $\mathcal{P}^{\text{count}}(A)$ as the set of all countable subsets of A. Prove or disprove each of the following statements:

(a) If A is countable, then $\mathcal{P}^{\text{fin}}(A)$ is countable.

(b) If A is countable, then $\mathcal{P}^{\text{count}}(A)$ is countable.

Problem 16. Let X be a countably infinite set and Y be a finite set with $|Y| \geq 2$. Prove or disprove each of the following statements:

(a) The set of all 1-1 functions from X to Y is uncountable.

(b) The set of all onto functions from X to Y is uncountable.

5 Counting

Counting was explored in the second half of Handout 3. This is essentially cardinality for finite sets. The main tool is counting argument (Tutorial 10), about counting a set of objects in two different ways (and thus concluding the two numbers are equal).

Problem 17. Refer to Tutorial 4 problem 13 for the definition of L-tile. Let n be a positive integer. How many ways are there to tile a $3 \times n$ board using L-tiles?

Problem 18. Let n be a positive integer. A **composition of n** is a way to write n as a sum of positive integers. The order matters. For example, $n = 3$ has 4 compositions: (3), $(2 + 1)$, $(1 + 2)$, and $(1 + 1 + 1)$.

How many compositions of n are there?

Let $A, B \subseteq \mathbb{Z}$. Their **sum-set**, denoted $A \oplus B$, is defined as the set of all possible sums of some element in A and some element in B, i.e.

$$A \oplus B = \{a + b \mid a \in A, b \in B\}.$$

(Also recall a similar problem from Quiz 2.)

Problem 19. Let A, B be finite subsets of $\mathbb{Z}$. Prove each of the following inequalities:

(a) $|A \oplus B| \leq |A| \cdot |B|$ (b) $|A \oplus B| \geq |A| + |B| - 1$

Problem 20. Let S be a finite subset of $\mathbb{Z}$ with $|S| = n$. Prove

$$|S \oplus S| \leq \frac{n(n+1)}{2}.$$

In other words, the inequality (a) in the previous problem can be tightened if you know $A = B$.

Let $f : X \to X$ be a function. A **fixed point** is an element $x \in X$ such that $f(x) = x$.

Problem 21. Let X be a finite set and $|X| = n$. How many functions $f : X \to X$ are there that have exactly 1 fixed point?

Problem 22. Let X be a finite set and $|X| = n$. Let $\pi : X \to X$ be a permutation. Prove π cannot have exactly $n - 1$ fixed points.

Problem 23. A function $f : X \to X$ is **idempotent** if $f \circ f = f$.

Let $|X| = n$. Prove the number of idempotent functions on X is given by the following sum:

$$\sum_{k=1}^{n} \binom{n}{k} k^{n-k}.$$

(**Hint:** Look at the fixed points of f.)

Problem 24. A function $f : X \to X$ is an **involution** if $f \circ f = i_X$ is the identity function.

Let A_n be the number of involutions on a set X with n elements. For $n \geq 3$, prove the following equation:

$$A_n = A_{n-1} + (n-1) \cdot A_{n-2}$$

6 Graph theory, properties you know

Graphs were defined in the middle of Handout 2 (directed graphs on page 15, undirected graphs on page 17), but they were explored more deeply in Handout 4. This section primarily relies on concepts you already know, either from the handouts or the tutorials. The next section explores other concepts in graph theory that are not discussed in this course.

Problem 25. Let A be a nonempty set, R be an equivalence relation on A, and G be the undirected graph representing R. Prove G is connected if and only if R is complete, i.e. xRy for all $x, y \in A$.

Problem 26. Refer to Tutorial 10 problem 10 for the definition of complement of a graph. Let $G = (V, E)$ be a loopless undirected graph with $|V| = 6$. Prove either G or $\overline{G}$ (or both) contains a cycle of length 3.

Problem 27. Let $G = (V, E)$ be a finite forest. Let C be the set of connected components of G. Prove

$$|E| = |V| - |C|.$$

Problem 28. Let $G = (V, E)$ be a finite loopless undirected graph. Suppose G contains (at least) 2 different spanning trees.

(a) Prove G is cyclic.

(b) Prove G contains at least 3 different spanning trees. (This means it's impossible for G to have *exactly* 2 different spanning trees.)

Problem 29. Let $T = (V, E)$ be a finite tree with $|V| = n$. Prove there exists a vertex $r \in V$, such that if we treat T as a rooted tree with r as the root, the height of T is at most $n/2$.

Problem 30. Let $T = (V, E)$ be a rooted tree. The **descendant relation of $\boldsymbol{T}$** is the binary relation R on the set V, such that xRy if and only if $x = y$ or y is a descendant of x.

Let T be a rooted tree and R be its descendant relation. Prove R is a partial order.

7 Graph theory, properties you don't know

This section develops new concepts in graph theory that you have not learned before. That means you need to understand the concept and apply it. While the "concept and application" structure appeared in a few earlier problems, you'll find a whole lot of it here.

I think graphs are wonderful with so many fascinating results, and I think the course doesn't show enough of them (perhaps due to time). So this section is a glimpse to what you might find if you decide to explore graphs more thoroughly.

Let $G = (V, E)$ be a finite loopless undirected graph. For a vertex v, the **degree of $\boldsymbol{v}$**, denoted $\deg(v)$, is the number of edges that contain v, i.e.

$$\deg(v) = |\{e \in E \mid v \in e\}|.$$

Problem 31. Let $G = (V, E)$ be a finite loopless undirected graph. Prove the degree sum formula:

$$\sum_{v \in V} \deg(v) = 2 \cdot |E|$$

Here the notation $\sum_{v \in V}$ means the sum over all vertices v in V.

Problem 32. Using the degree sum formula (Problem 31) or otherwise, prove each of the following statements:

(a) If $G = (V, E)$ is a finite tree and $|V| \geq 2$, there exist at least 2 vertices of degree 1.

(b) In any finite loopless undirected graph G, there exist an even number of vertices whose degree is odd. (This result is called the "handshaking lemma".)

Problem 33. Let $G = (V, E)$ be a finite loopless undirected graph. If $|V| \geq 2$, prove there exist 2 vertices of the same degree.

> Let $G = (V, E)$ be a finite loopless undirected graph. For natural number k, a ***k*-coloring of *G*** is a function $f : V \to \{1, 2, \ldots, k\}$ such that, whenever $\{u, v\} \in E$, we have $f(u) \neq f(v)$. In other words, adjacent vertices never get the same color. If a graph G has a k-coloring, we say G is ***k*-colorable**.

Problem 34. Let $G = (V, E)$ be a finite loopless undirected graph. Let d be a natural number such that $\deg(v) \leq d$ for all $v \in V$. Prove G is $(d + 1)$-colorable.

Problem 35. Let $G = (V, E)$ be a finite loopless undirected graph. Prove G is 2-colorable if and only if it does **not** contain a cycle with an odd number of vertices.

Problem 36. Let $G = (V, E)$ be a finite tree with n vertices. Let $k \geq 2$ be an integer. How many k-colorings of G are there?

Problem 37. Let $G = (V, E)$ be a finite loopless undirected graph. Suppose that G has exactly k connected components. Prove that the number of 2-colorings of G is either 0 or 2^k.

> Let $G = (V, E)$ be a finite connected undirected graph. The **distance** between two vertices x, y, denoted $\mathrm{dist}_G(x, y)$, is the minimum length among all paths between x and y in G. For convenience, it is useful to define $\mathrm{dist}_G(x, x) = 0$.
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> The **diameter of *G***, denoted $\mathrm{diam}(G)$, is the maximum distance between any pair of vertices in G.

Problem 38. Let $G = (V, E)$ be a finite connected undirected graph. Prove the triangle inequality: for all $x, y, z \in V$,

$$\mathrm{dist}_G(x, z) \leq \mathrm{dist}_G(x, y) + \mathrm{dist}_G(y, z).$$

Problem 39. Let $G = (V, E)$ be a finite connected undirected graph with $|V| = n$. Prove $\mathrm{diam}(G) = n - 1$ if and only if G is a path.

Problem 40. Let $T = (V, E)$ be a finite rooted tree with height h. Prove

$$\mathrm{diam}(T) \leq 2h.$$